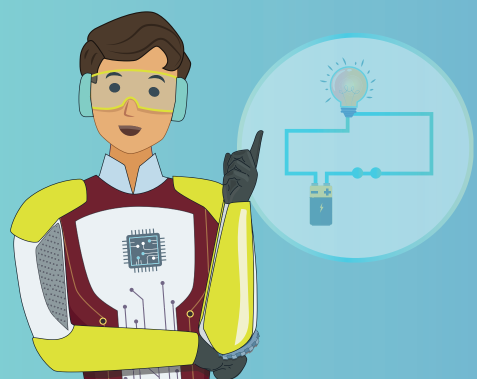
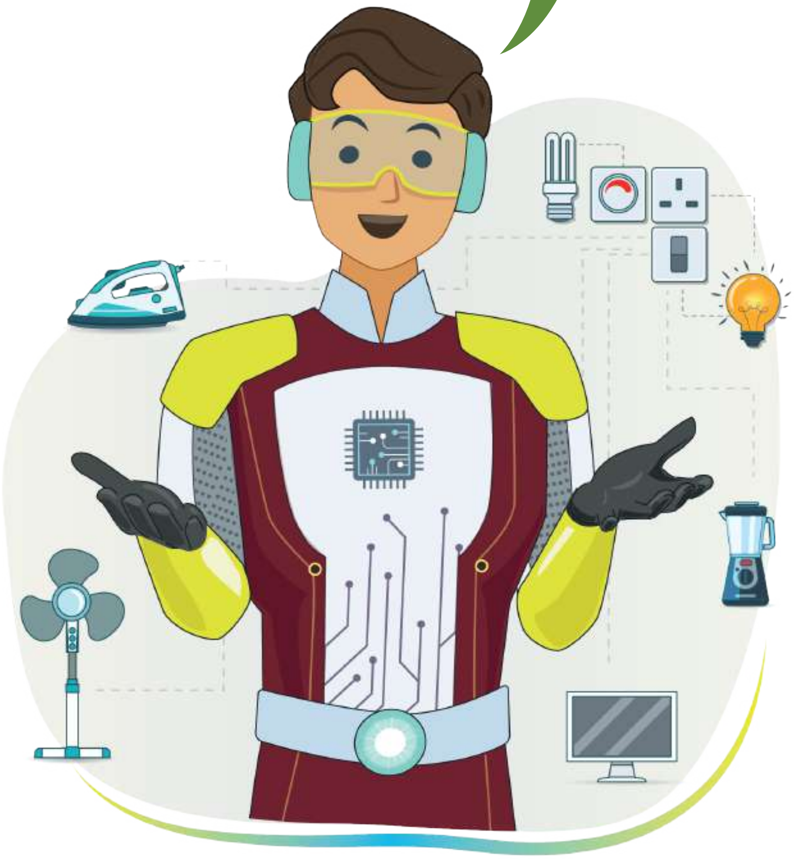


Electronic Circuits

15

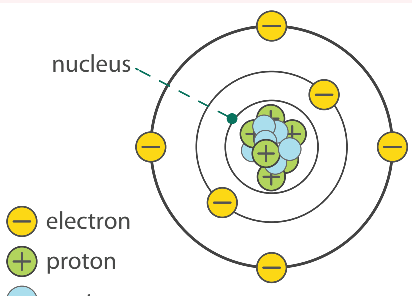
Isn't it amazing how all of our devices use electricity to operate? Let us see how electricity runs through a circuit and how we can measure it.



All electronic devices operate via an electric current. Electricity provides this current and passes it through devices via charged particles.

Each material has two types of charged particles:

- Protons as positively charged particles.
- Electrons as negatively charged particles.



Particles

Four concepts combine to make an electronic device work:

1. Electrons
2. Electric Current
3. Voltage
4. Resistance

Recall:

1. Electrons

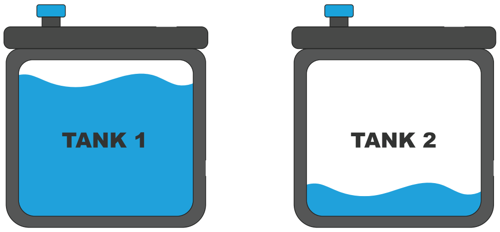
Electrons are negatively charged subatomic particles. They can either be free or attached to the nucleus in an atom.

2. Electric Current

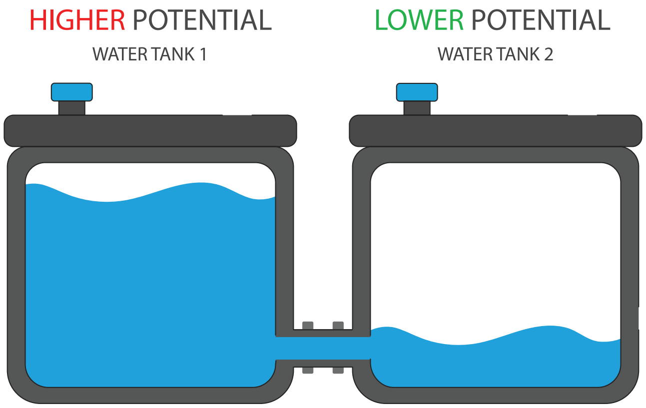
Electric current refers to the flow of electrons in a circuit. Higher the number of electrons, higher the current. Electric current flows through a wire and is measured in Amperes (A).

3. Voltage

Imagine that there are two water tanks. Tank 1 is almost full, and Tank 2 is almost empty. What will happen if we connect both tanks with a pipe?



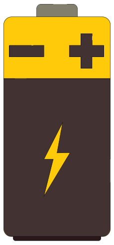
There is a difference in the water level in both tanks. This will cause the water from Tank 1 to flow to Tank 2 until the water level is equal in both tanks. This is due to the pressure generated by the huge amount of water in Tank 1. We say that Tank 1 has a higher potential than Tank 2. Water will always move from a higher potential to a lower potential.



Similarly, if we replace the water tanks with a battery and the pipe with a wire, then the potential difference in the level will be similar to the voltage.

Voltage is the pressure from an electric force that pushes charged electrons from one point to another. It is measured in volts (V).

Voltage pushes electrons in the wire to flow from the battery's negative point to its positive point. A battery is similar to a container that stores energy until it is needed.

Battery:  Battery symbol: 

Since the current increases with an increase in voltage, it is important to control current flow.

For example:

Water stored in a dam is energy with huge pressure. If it is released without control, it can cause tremendous destruction. The gates of the dam control it, and it is released as required.

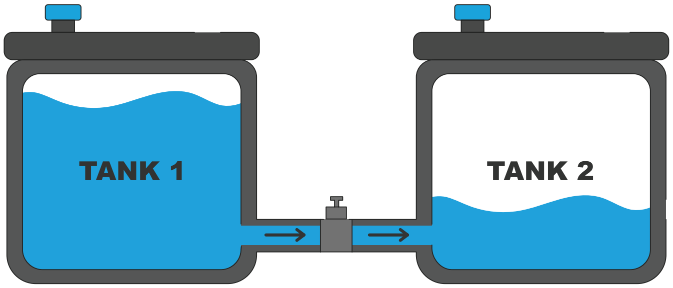
Similarly, resistance is used to control the flow of current in a circuit.

4. Resistance

Resistance is a phenomenon in a material that controls the current flowing through it. Resistance is derived from the phrase 'to resist' which means 'to hold'. This makes resistance inversely proportional to the flow of current.

Higher the resistance, lower the current flow.

For example: In the above water tank example, add a valve to the pipe. That will act as a resistance and control the flow of water through the pipe.



15. Electronic Circuits

The value of a resistor is measured in Ohm (Ω).

Resistor: 

Symbol: 

We have seen that the amount of electric current flowing through a wire depends on the values of voltage and resistance. This value can be calculated using Ohm's Law. It is one of the most important laws of electric circuits.

Ohm's Law

Ohm's law states that the current flowing in a circuit is directly proportional to the voltage and inversely proportional to the resistance. That means the electric current will increase with an increase in voltage, and decrease if the resistance increases.

The symbol for ohm is "Ω".

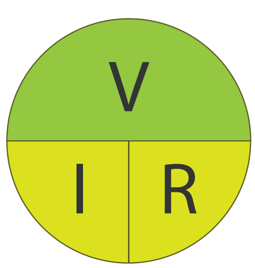
Formula to calculate electric current

I=V/R

Where,

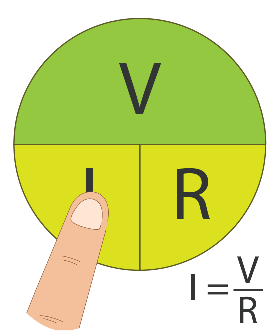
I is electric current **V** is voltage **R** is resistance

We can easily calculate the values of all the components in a circuit with the circle of Ohm's Law.



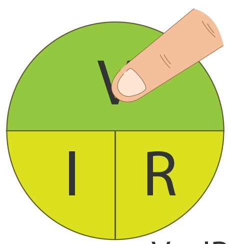
Circle of Ohm's Law

When we need to find I (Electric Current)



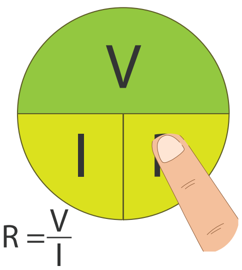
15. Electronic Circuits

When we need to find V (Voltage)



$V = IR$

When we need to find R (Resistance)

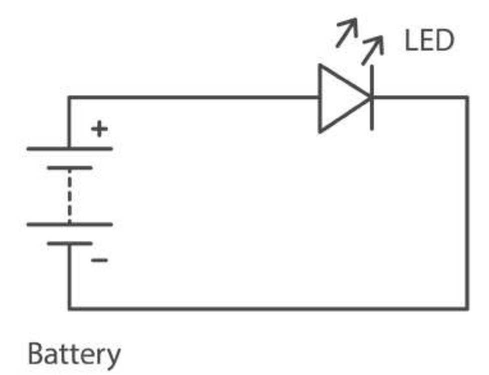


$R = \frac{V}{I}$

Ohms law can be applied when electronic components are connected in a circuit.

Circuit

A circuit is a closed path that allows electric current to flow from one point to another. This closed path may include various electronic components, such as a battery, resistors, LED, etc.



There are two types of circuits:

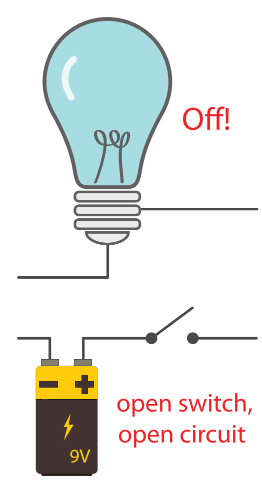
a. Open Circuits

b. Closed Circuits

Open circuit

When the path of a current is interrupted due to a gap or insulator, it cannot complete the flow from the positive to the negative terminal. This kind of path is called an open circuit.

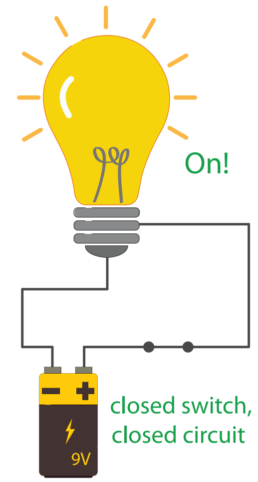
For example: When the switch is turned OFF, the LED bulb will go off.

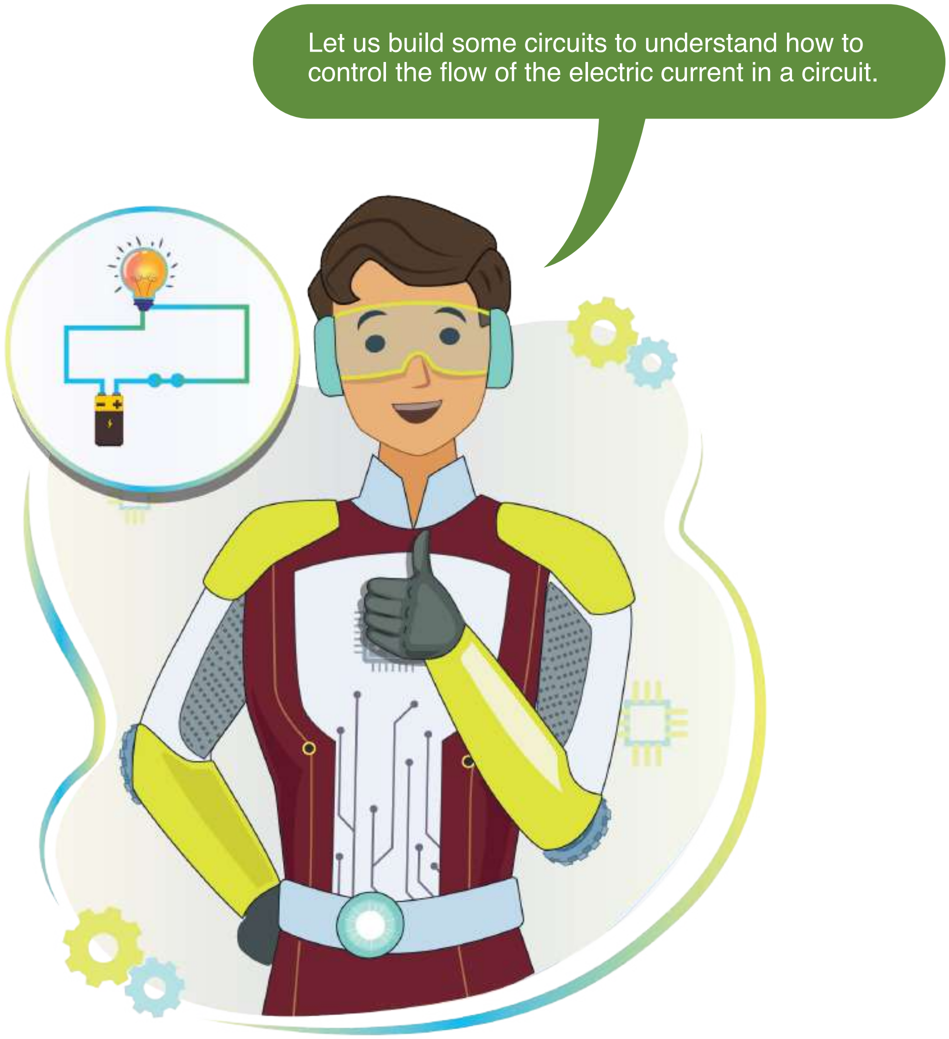


Closed circuit

When the path of a current is not interrupted with any gap or insulator, it can complete its flow from the positive to the negative terminal. This kind of path is called a closed circuit.

For example: When the switch is turned ON, the LED bulb will go ON.



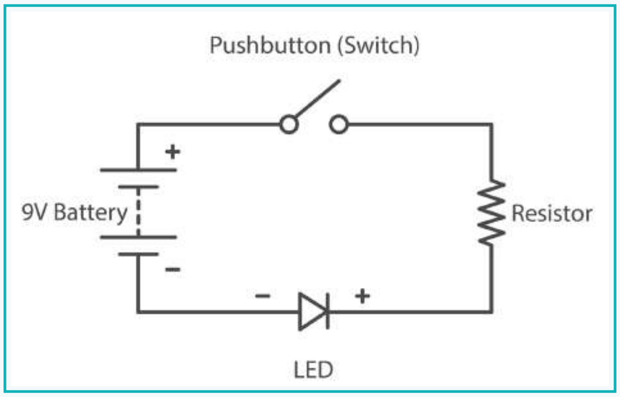


Activity

- Create a circuit for LED ON and OFF with Tinkercad circuit. Test how much current can be passed through an LED without damaging it.

  Open Tinkercad circuit on the Chrome browser. Join the class or Sign In.

Let us first understand the circuit diagram.



Here, the battery is working as a source of energy. It is used to provide a current to the LED. A resistor is connected to resist the excess current flow, and a switch is connected to close or open the circuit.

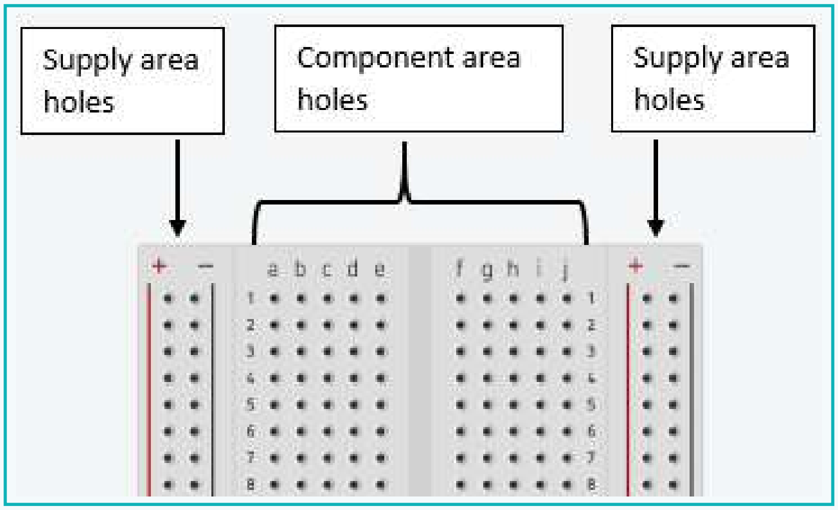
Let us create a circuit on Tinkercad to switch the LED ON and OFF.

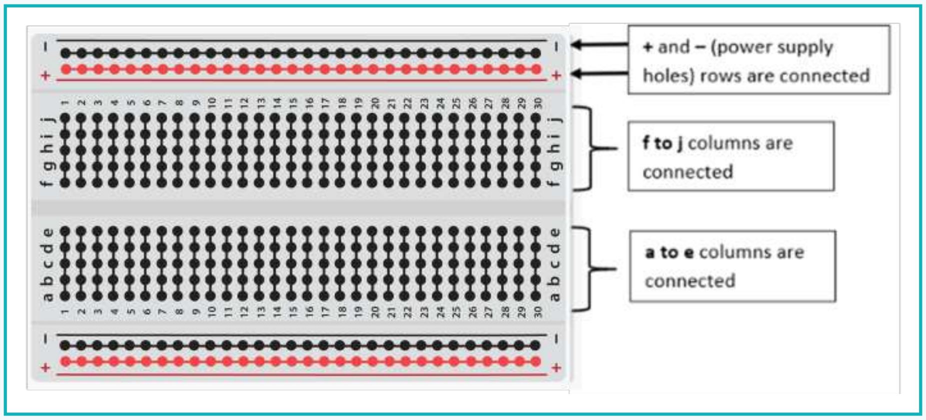
A breadboard can be used to connect components and create a circuit.

Activity

Breadboard: It is used to connect more than one component at the same time. Breadboards have holes to stick components on them to create the circuits.

A breadboard has four sections: two for the power supply area and two for component connections.





How do we use a breadboard?

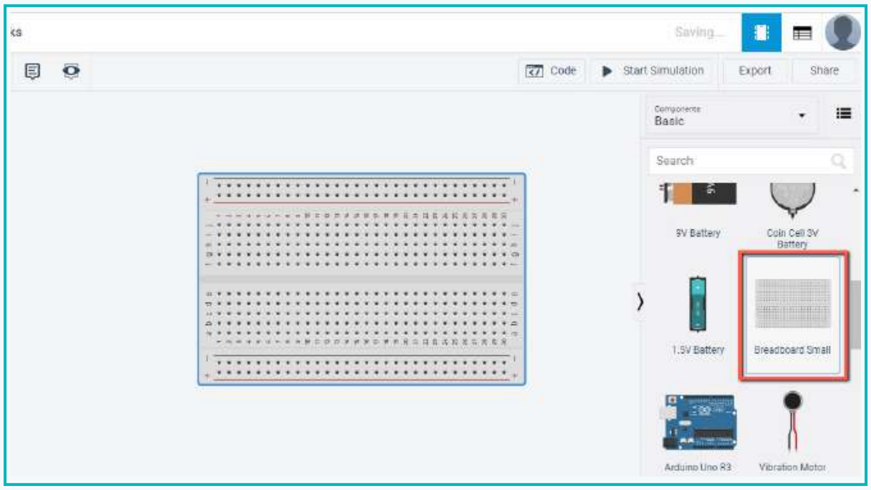
a. To provide supply or to connect any component, we use vertical holes.

b. We cannot connect components in horizontal holes with each other.

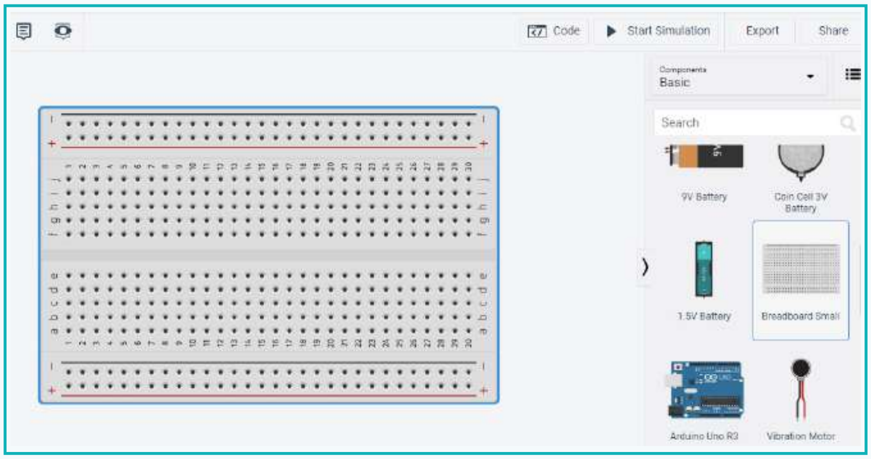
c. We provide power supply to all the components by connecting wires to supply area holes.

Activity

1 Drag and drop the breadboard



- We can scroll to increase or decrease the size of a component.

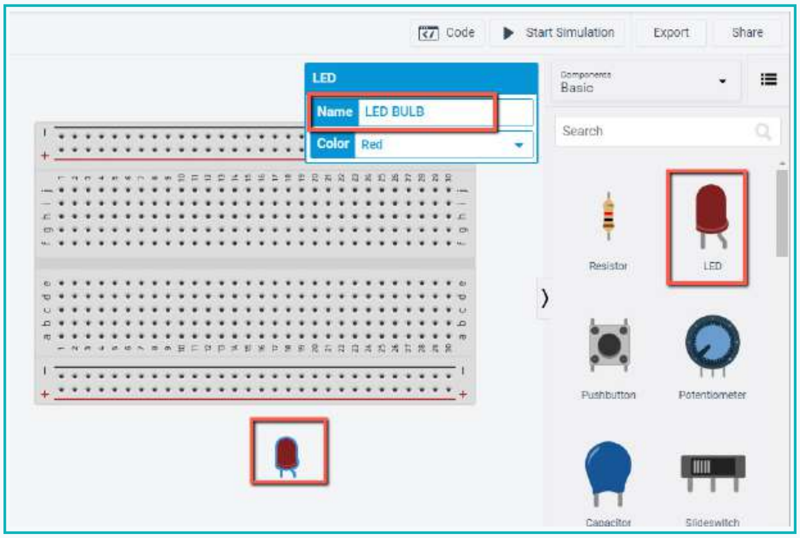


Here, LED will work as an LED bulb and Pushbutton will work as an ON-OFF switch.

Activity

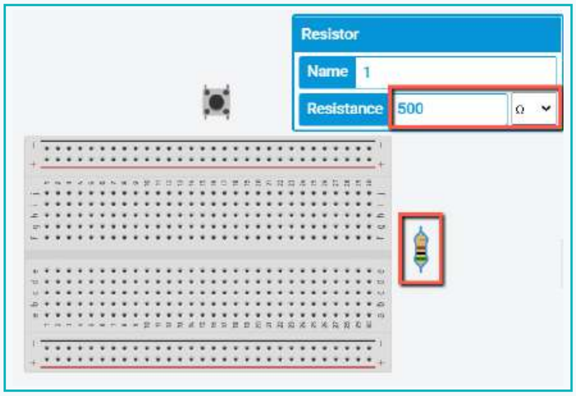
2 Drag LED and change its name to "LED BULB"

- Choose any colour for the LED.

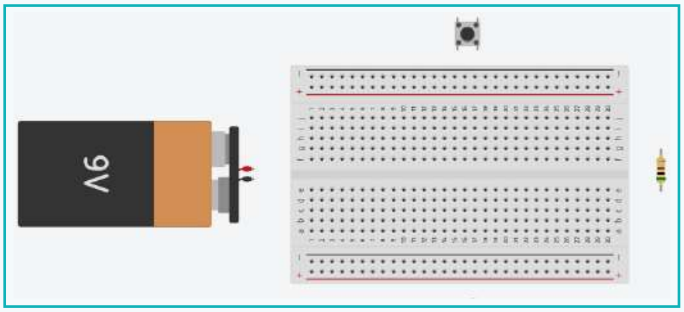


3 Drag Pushbutton and change its name to "Switch"

- Drag a resistor of 500ohms (change the value) and a battery in the same way.

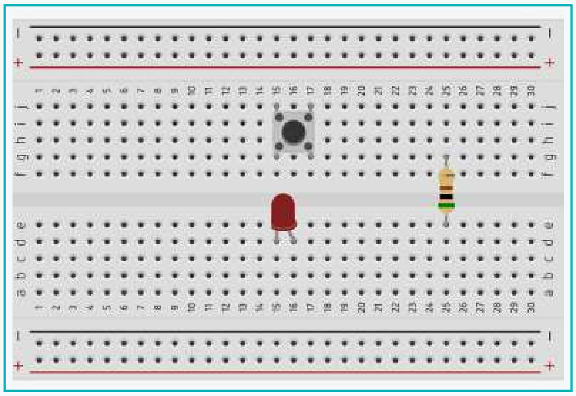


4 To rotate the battery, use the rotate option

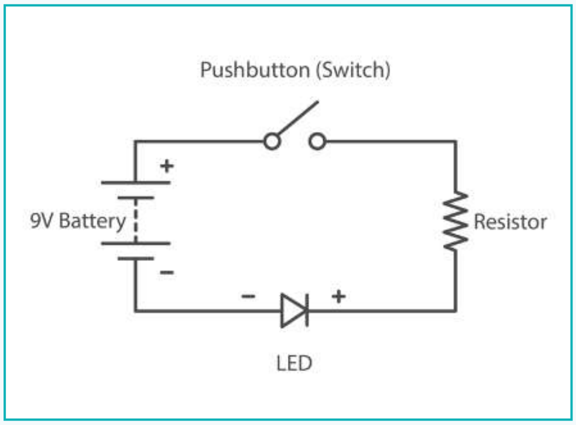


Activity

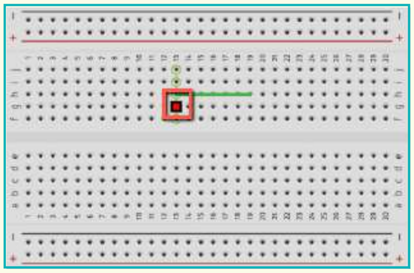
5 Place the component on the breadboard as shown below (use drag and drop)



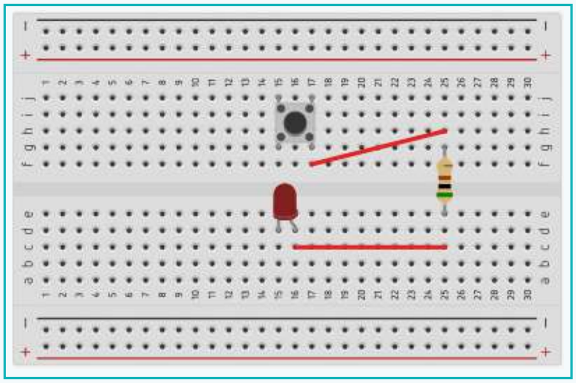
6 Observe the circuit diagram and connect components according to the diagram



Note:
To connect any component, we need a wire. Drag the cursor from one hole to another to connect them.



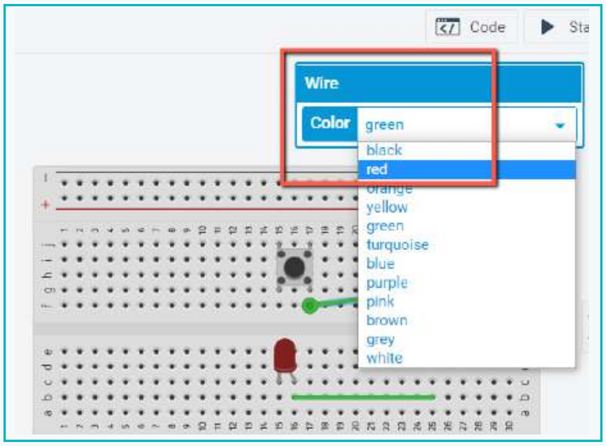
7 Connect one terminal of the resistor to the LED, and the other to the pushbutton as shown



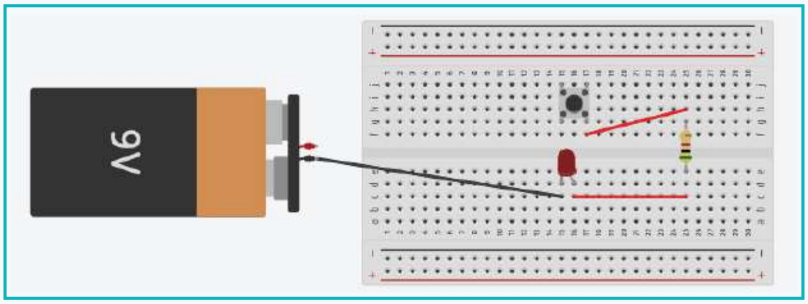
Activity

Use two different colours to indicate positive and negative connections.

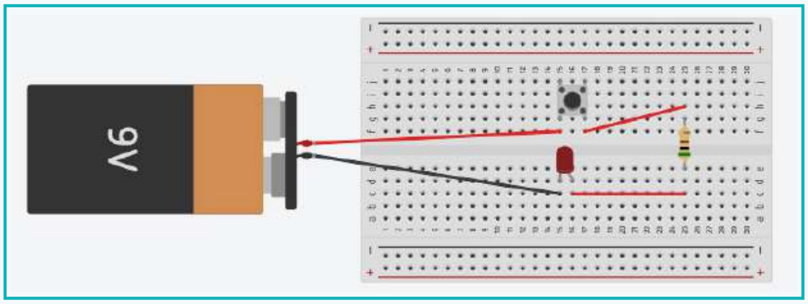
8 Select the wire and change its colour to red when it connects to the positive terminal



9 Move the cursor on the component terminals to view the pin details. Connect the negative terminal of the LED to the battery's negative terminal using a black wire



10 Connect one terminal of the pushbutton to the battery's positive terminal



In this circuit, when we press the pushbutton, the LED will turn ON. When we release the pushbutton, the LED will go OFF.

Activity

We know the formula of Ohm's law, let us use it to calculate the value of the current.

The LED needs 0.020amp to light up, which is safe for an LED.

The formula of Ohm's law is "I = V/R"

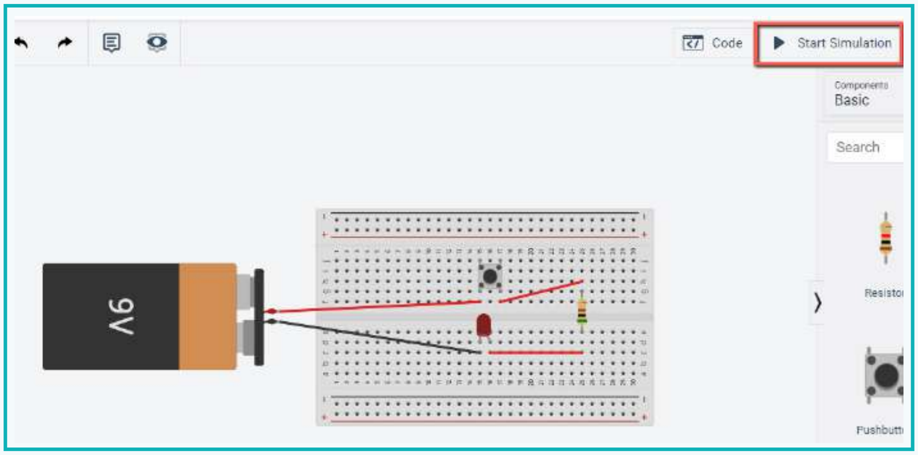
We have a battery of 9V, therefore voltage V = 9V

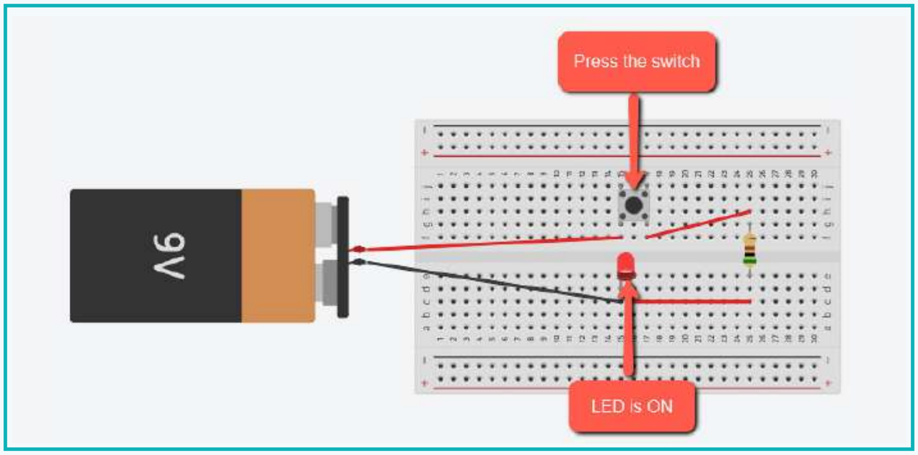
We are using 500 Ω resistance, so R = 500 Ω

I = 9/500

I = 0.018amp (Ampere is the unit of current)

11 To check the output, click on **start simulation** and click on the pushbutton

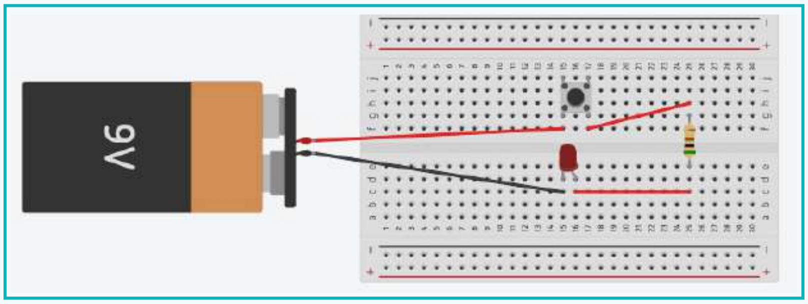




LED is ON

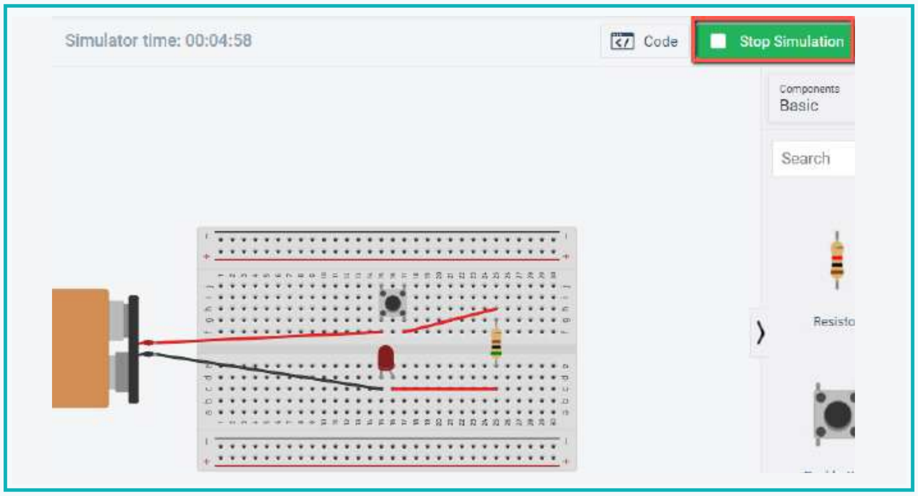
Activity

12 Release the pushbutton to switch OFF the LED

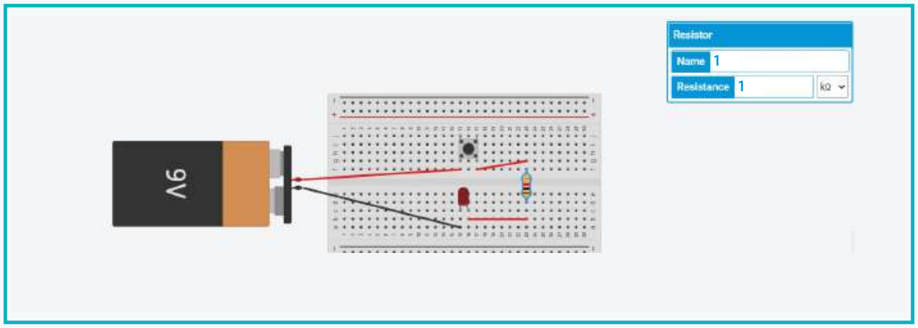


LED is OFF

13 Click on **stop simulation** to stop the circuit



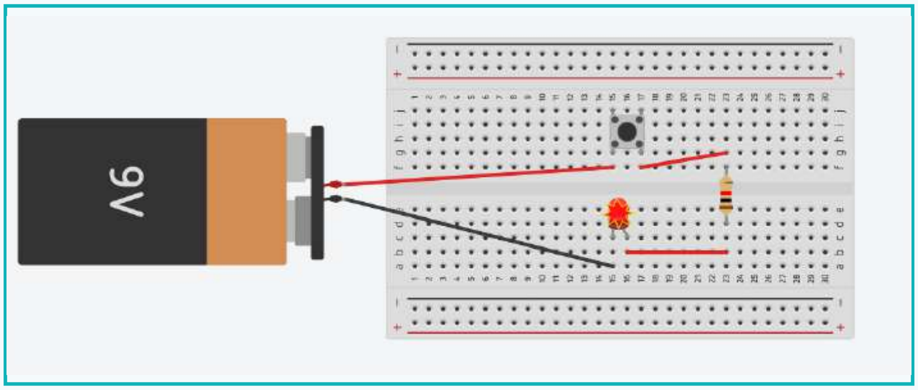
14 Change the value of the resistor to 1 KΩ



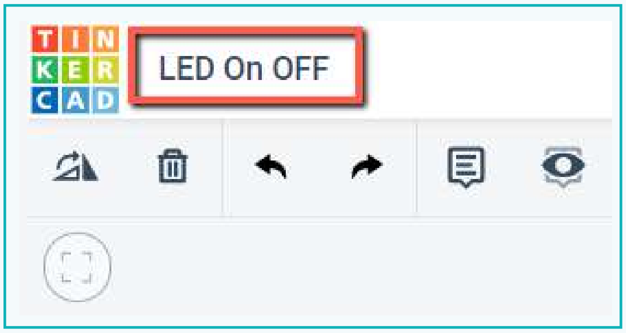
Activity

15 Start the simulation and check the LED light. Because of the high resistance, the LED will not light up or the light's intensity will be less

16 If the resistance value is set to 0 or close to 0, then a high current will flow through the circuit and it will damage the LED

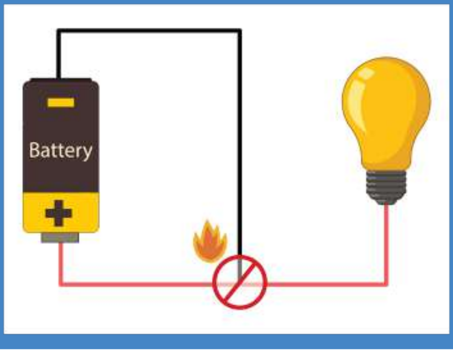


17 Change the project name to "LED On OFF"



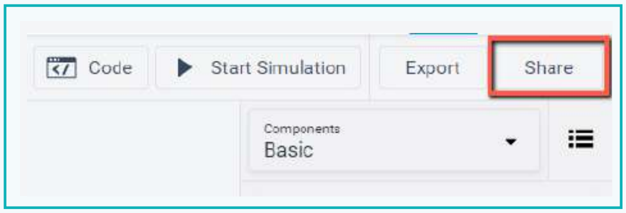
Bytes

Short circuit refers to an unwanted connection between two terminals that causes a problem in a circuit. It can also allow a very high current flow that can destroy the components or create sparks/fire. Always be cautious and recheck the circuit before connecting it with the energy source.



Activity

18 Click on Share to download the circuit.



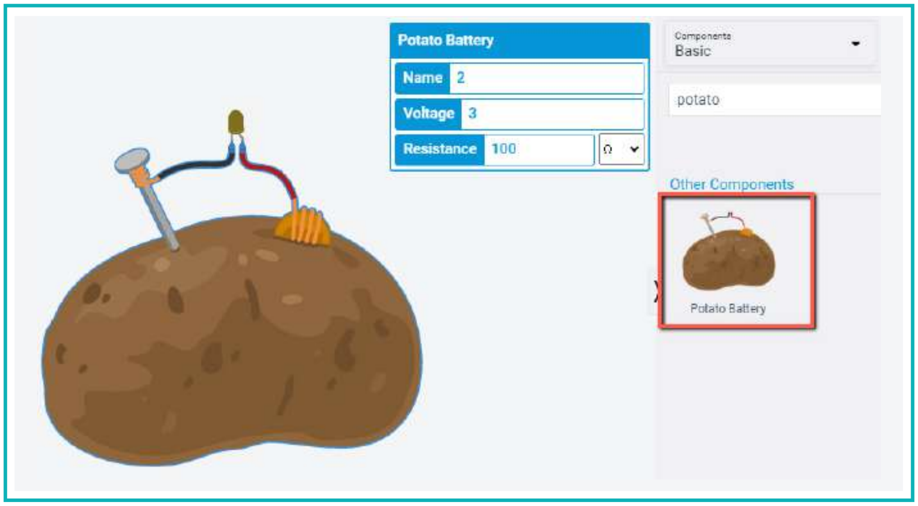
19 Check for the circuit in the downloads folder.

Try:

a. A potato can be used as a battery to provide enough energy to light up an LED.

b. Add a potato battery from basic components, and change its properties to Voltage = 3V and Resistance = 100 Ω.

c. Connect it with an LED and run the simulation.



Exercise

State whether True or False

1. Resistor is not connected in circuit with LED with 9v battery voltage then, then LED will be safe.

▲ ____________________

2. The formula to calculate voltage is V= IR.

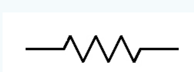
▲ ____________________

Tick mark ☑ the correct answer

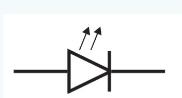
3. Components that can be used as a switch are:

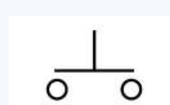
a. LED ☐ **b.** Breadboard ☐
c. Resistor ☐ **d.** Pushbutton ☐

4. Identify the symbol of a resistor from the following:

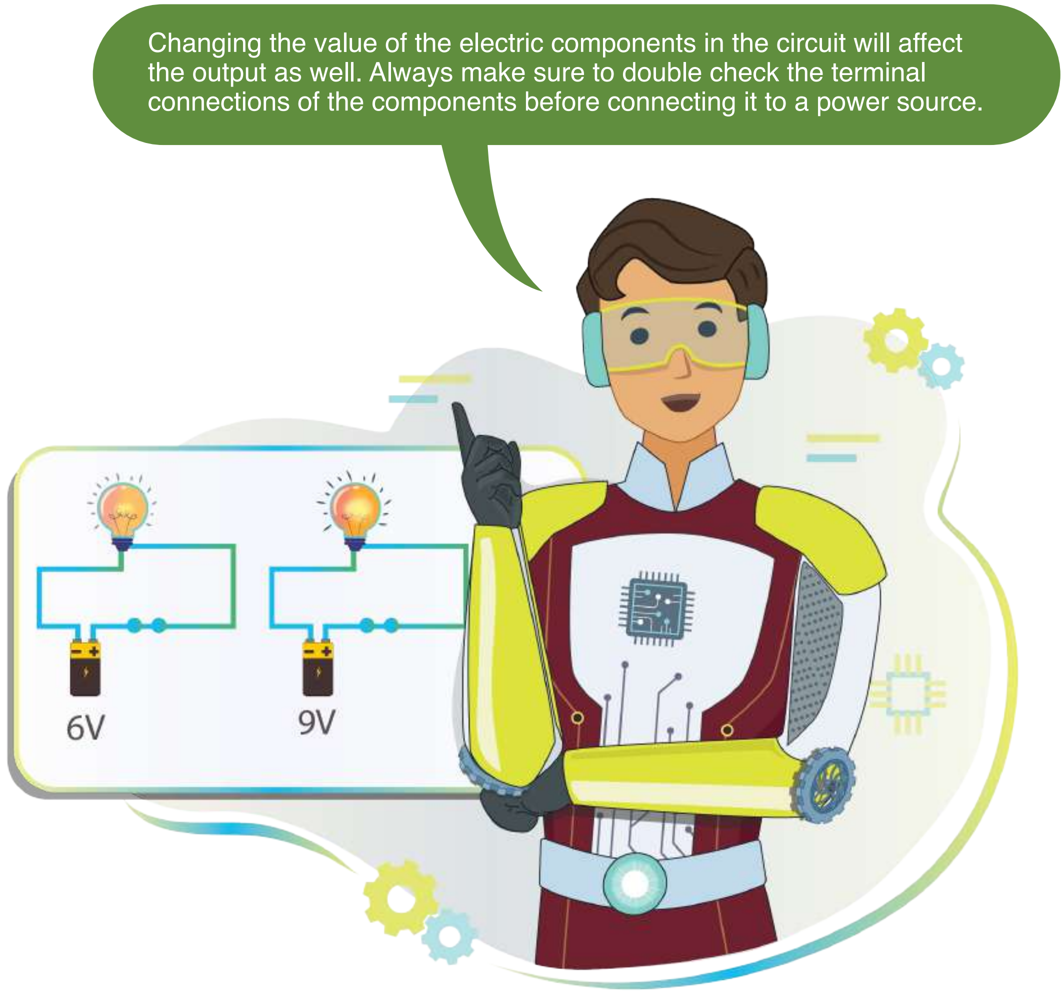
a. ☐ 

b. ☐ 

c. ☐ 

d. ☐ 

Ohm's law can be used to find the values of resistance, voltage and electric current. This helps us determine the values that we need to create a working electric circuit. A working electric circuit will always be a closed-circuit as the electric current always flows from the positive to the negative terminal in a closed circuit. When the circuit is open, the electric current will not flow through it.



Tech Flash

- **Tinkercad** : Tinkercad is an online tool to design electronic circuits.
- **Voltage** : Voltage is an electric force that causes electrons to move from one point to another.
- **Resistance** : Resistance is a phenomenon of a material that limits the current flowing through it.
- **Ohm's Law** : Ohm's law states that electric current is directly proportional to voltage, and inversely proportional to resistance.
- **Circuit** : Circuit is a rough path that allows electricity to flow.
- **Battery** : Battery is used to provide power to electronic components.
- **LED** : LED is an electronic device that releases light when an electrical current is passed through it.
- **Pushbutton** : The pushbutton is a component that connects two points in a circuit when pressed.